

1. Database Setup & Data Entry

- Create a new database. ● Create the three tables with proper: ○ Primary keys, Foreign keys, Appropriate data types ● Insert sample data: ○ 4–5 learners ○ 4–5 courses (different categories) ○ 6–8 purchase records

2. Data Exploration Using Joins Use INNER, LEFT, and RIGHT JOIN to: ● Combine learner, course, and purchase data ● Display: Learner name, Course name, Category, Quantity, Total amount, Purchase date ● Presentation Requirements: ■ Format currency to 2 decimal places ■ Use column aliases ■ Sort by the highest total amount

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
112 RIGHT JOIN courses c ON p.purchase_id = c.purchase_id
113 LEFT JOIN learners l ON p.learner_id = l.learner_id
114 ORDER BY (p.quantity * c.unit_price) DESC;
```

The query is executed, and the results are displayed in the Result Grid. The results are sorted by total amount in descending order.

learner_name	course_name	category	quantity	total_amount	purchase_date
Alice Johnson	Advanced Machine Learning	Advanced	1	15,000.00	2024-02-10
Sofia Rossi	Advanced Machine Learning	Advanced	1	15,000.00	2024-03-15
Rahul Mehta	Digital Marketing Fundamentals	Intermediate	2	10,000.00	2024-03-05
Sofia Rossi	Python for Beginners	Beginner	3	6,000.00	2024-01-30
Chen Wei	Digital Marketing Fundamentals	Intermediate	1	5,000.00	2024-02-25
Rahul Mehta	Excel Basics	Beginner	2	2,400.00	2024-01-20
Alice Johnson	Python for Beginners	Beginner	1	2,000.00	2024-01-15
John Smith	Excel Basics	Beginner	1	1,200.00	2024-02-05

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
97 p.purchase_date AS purchase_date
98 FROM learners l
99 LEFT JOIN purchases p ON l.learner_id = p.learner_id
100 LEFT JOIN courses c ON p.purchase_id = c.purchase_id
101 ORDER BY (p.quantity * c.unit_price) DESC;
```

The query is executed, and the results are displayed in the Result Grid. The results are sorted by total amount in descending order.

learner_name	course_name	category	quantity	total_amount	purchase_date
Alice Johnson	Advanced Machine Learning	Advanced	1	15,000.00	2024-02-10
Sofia Rossi	Advanced Machine Learning	Advanced	1	15,000.00	2024-03-15
Rahul Mehta	Digital Marketing Fundamentals	Intermediate	2	10,000.00	2024-03-05
Sofia Rossi	Python for Beginners	Beginner	3	6,000.00	2024-01-30
Chen Wei	Digital Marketing Fundamentals	Intermediate	1	5,000.00	2024-02-25
Rahul Mehta	Excel Basics	Beginner	2	2,400.00	2024-01-20
Alice Johnson	Python for Beginners	Beginner	1	2,000.00	2024-01-15
John Smith	Excel Basics	Beginner	1	1,200.00	2024-02-05

MySQL Workbench interface showing a SQL query and its results. The query is as follows:

```

82 p.quantity AS quantity,
83 FORMAT(p.quantity * c.unit_price, 2) AS total_amount,
84 p.purchase_date AS purchase_date
85 FROM purchases p
86 INNER JOIN learners l ON p.learner_id = l.learner_id
87 INNER JOIN courses c ON p.course_id = c.course_id
88 ORDER BY (p.quantity * c.unit_price) DESC
89

```

The results table displays the following data:

learner_name	course_name	category	quantity	total_amount	purchase_date
Alice Johnson	Advanced Machine Learning	Advanced	1	15,000.00	2024-02-10
Sofia Rossi	Advanced Machine Learning	Advanced	1	15,000.00	2024-03-15
Rahul Mehta	Digital Marketing Fundamentals	Intermediate	2	10,000.00	2024-03-05
Sofia Rossi	Python for Beginners	Beginner	3	6,000.00	2024-01-30
Chen Wei	Digital Marketing Fundamentals	Intermediate	1	5,000.00	2024-02-25
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Alice Johnson	Python for Beginners	Beginner	1	2,000.00	2024-01-15
John Smith	Excel Basics	Beginner	1	1,200.00	2024-02-05

The Action Output section shows the following messages:

- 23 16:26:16 WITH leamer_spending AS (SELECT learner_id, l.full_name, S... 5 row(s) returned
- 24 16:26:31 WITH leamer_spending AS (SELECT learner_id, l.full_name, S... 5 row(s) returned
- 25 16:40:19 SELECT l.full_name AS leamer_name, c.course_name ... 8 row(s) returned

3. Core Analytical Queries (Q1–Q5) Q1. Display each learner’s total spending with their country. Q2. Find the top 3 most purchased courses by quantity. Q3. Show each category’s: ● Total revenue ● Number of unique learners

Q4. List learners who purchased from more than one category. Q5. Identify courses never purchased.

MySQL Workbench interface showing a SQL query and its results. The query is as follows:

```

166 c.course_id AS course_id,
167 c.course_name AS course_name,
168 c.category AS category
169 FROM courses c
170 LEFT JOIN purchases p ON c.course_id = p.course_id
171 WHERE p.purchase_id IS NULL
172
173 -- SECTION 4: SUBQUERIES & CORRELATED SUBQUERIES (Q6-Q8)

```

The results table displays the following data:

course_id	course_name	category
105	Cloud Architecture Mastery	Expert

The Action Output section shows the following messages:

- 30 16:42:43 SELECT c.category AS category, FORMAT(SUM(p.quantity * c.unit_price), 2) ... 3 row(s) returned
- 31 16:42:48 SELECT l.full_name AS leamer_name, COUNT(DISTINCT c.category) AS cat... 3 row(s) returned
- 32 16:42:55 SELECT c.course_id AS course_id, c.course_name AS course_name, c.c... 1 row(s) returned

MySQL Workbench

unconnected x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- analysis_db
- analyst_db
- books_db
- college_database
- companydb
- e_platform
- employee
- employee_db
- employee_dbe

Administration Schemas

Information

No object selected

SQL File 4* SQL Day 1 CCSCRIPT* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 9* SQL_final_mo

Limit to 2000 rows

```

157 FROM learners l
158 JOIN purchases p ON l.learner_id = p.learner_id
159 JOIN courses c ON p.course_id = c.course_id
160 GROUP BY l.learner_id, l.full_name
161 HAVING COUNT(DISTINCT c.category) > 1
162 ORDER BY category_count DESC;
163
164 -- ---- Q5: Courses never purchased ----

```

Result Grid

learner_name	category_count
Alice Johnson	2
Rahul Mehta	2
Sofia Rossi	2

Result 21 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
29	16:42:29	SELECT c.course_name AS course_name, c.category AS category, SUM...	3 row(s) returned	0.000 sec / 0.000 sec
30	16:42:43	SELECT c.category AS category, FORMAT(SUM(p.quantity * c.unit_price), 2) ...	3 row(s) returned	0.015 sec / 0.000 sec
31	16:42:48	SELECT l.full_name AS learner_name, COUNT(DISTINCT c.category) AS cat...	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

MySQL Workbench

unconnected x Local instance MySQL80 x

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Administration Schemas

Information

No object selected

SQL File 4* SQL Day 1 CCSCRIPT* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 9* SQL_final_mo

Limit to 2000 rows

```

148 FROM purchases p
149 JOIN courses c ON p.course_id = c.course_id
150 GROUP BY c.category
151 ORDER BY SUM(p.quantity * c.unit_price) DESC;
152
153 -- ---- Q4: Learners who purchased from more than one category ----
154 SELECT
155     l.full_name AS learner_name,

```

Result Grid

category	total_revenue	unique_learners
Advanced	30,000.00	2
Intermediate	15,000.00	2
Beginner	11,600.00	4

Result 20 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
28	16:42:22	SELECT l.full_name AS learner_name, l.country AS country, FORMAT(SU...	5 row(s) returned	0.000 sec / 0.000 sec
29	16:42:29	SELECT c.course_name AS course_name, c.category AS category, SUM...	3 row(s) returned	0.000 sec / 0.000 sec
30	16:42:43	SELECT c.category AS category, FORMAT(SUM(p.quantity * c.unit_price), 2) ...	3 row(s) returned	0.015 sec / 0.000 sec

Query Completed

MySQL Workbench

unconnected x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

analysis_db
analyst_db
books_db
college_database
companydb
e_platform
employee
employee_db
employee_dbe

Administration Schemas

Information

No object selected

SQL File 4* SQL Day 1 CCSCRIPT* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 9* SQL_final_ms

Limit to 2000 rows

SQL

```

136 SUM(p.quantity) AS total_quantity_sold
137 FROM purchases p
138 JOIN courses c ON p.course_id = c.course_id
139 GROUP BY c.course_id, c.course_name, c.category
140 ORDER BY total_quantity_sold DESC
141
142
143 -- ---- Q3: Each category's total revenue and number of unique learners ----

```

Result Grid

course_name	category	total_quantity_sold
Python for Beginners	Beginner	4
Excel Basics	Beginner	3
Digital Marketing Fundamentals	Intermediate	3

Result 19 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
27	16:40:45	SELECT l.full_name AS learner_name, c.course_name	9 row(s) returned	0.000 sec / 0.000 sec
28	16:42:22	SELECT l.full_name AS learner_name, l.country AS country, FORMAT(SU...	5 row(s) returned	0.000 sec / 0.000 sec
29	16:42:29	SELECT c.course_name AS course_name, c.category AS category, SUM...	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

MySQL Workbench

unconnected x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

analysis_db
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Administration Schemas

Information

No object selected

SQL File 4* SQL Day 1 CCSCRIPT* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 9* SQL_final_ms

Limit to 2000 rows

SQL

```

127 JOIN purchases p ON l.learner_id = p.learner_id
128 JOIN courses c ON p.course_id = c.course_id
129 GROUP BY l.learner_id, l.full_name, l.country
130 ORDER BY SUM(p.quantity * c.unit_price) DESC;
131
132 -- ---- Q2: Top 3 most purchased courses by quantity ----
133 SELECT
134 c.course_name AS course_name,

```

Result Grid

learner_name	country	total_spending
Sofia Rossi	Italy	21,000.00
Alice Johnson	USA	17,000.00
Rahul Mehta	India	12,400.00
Chen Wei	China	5,000.00
John Smith	USA	1,200.00

Result 18 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
26	16:40:38	SELECT l.full_name AS learner_name, c.course_name	9 row(s) returned	0.000 sec / 0.000 sec
27	16:40:45	SELECT l.full_name AS learner_name, c.course_name	9 row(s) returned	0.000 sec / 0.000 sec
28	16:42:22	SELECT l.full_name AS learner_name, l.country AS country, FORMAT(SU...	5 row(s) returned	0.000 sec / 0.000 sec

Query Completed

4. Subqueries & Correlated Subqueries Q6. Find learners whose total spending is above the average learner spending. Q7. Display courses whose price is higher than any course in the 'Beginner' category. Q8 . Find learners who spent more than the average spending in their country.

MySQL Workbench

unconnected x Local instance MySQL80 x

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Administration Schemas

Information

No object selected

SQL File 4* SQL Day 1 CCSCRIPT* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 9* SQL_final_mo

Limit to 2000 rows

```

223 JOIN purchases p2 ON l2.learner_id = p2.learner_id
224 JOIN courses c2 ON p2.course_id = c2.course_id
225 WHERE l2.country = 1.country
226 GROUP BY l2.learner_id
227 ) AS country_avg
228
229 ORDER BY total_spending DESC;
230

```

Result Grid

learner_name	country	total_spending
Alice Johnson	USA	17,000.00

Result 25 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
33	16:45:33	SELECT l.full_name AS learner_name, FORMAT(l.total_spending, 2) AS total...	3 row(s) returned	0.000 sec / 0.000 sec
34	16:45:41	SELECT course_name AS course_name, category AS category, FORMA...	4 row(s) returned	0.016 sec / 0.000 sec
35	16:45:48	SELECT l.full_name AS learner_name, l.country AS country, FORMAT(SU...	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

unconnected x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

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Administration Schemas

Information

No object selected

SQL File 4* SQL Day 1 CCSCRIPT* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 9* SQL_final_mo

Limit to 2000 rows

```

205 SELECT unit_price FROM courses WHERE category = 'Beginner'
206
207 ORDER BY unit_price DESC;
208
209 -- ---- Q8: Learners who spent more than the average spending in their own country ----
210 SELECT
211     l.full_name AS learner_name,
212     l.country AS country,

```

Result Grid

course_name	category	unit_price
Digital Marketing Fundamentals	Intermediate	5,000.00
Python for Beginners	Beginner	2,000.00
Cloud Architecture Mastery	Expert	18,000.00
Advanced Machine Learning	Advanced	15,000.00

Result 24 x

Output

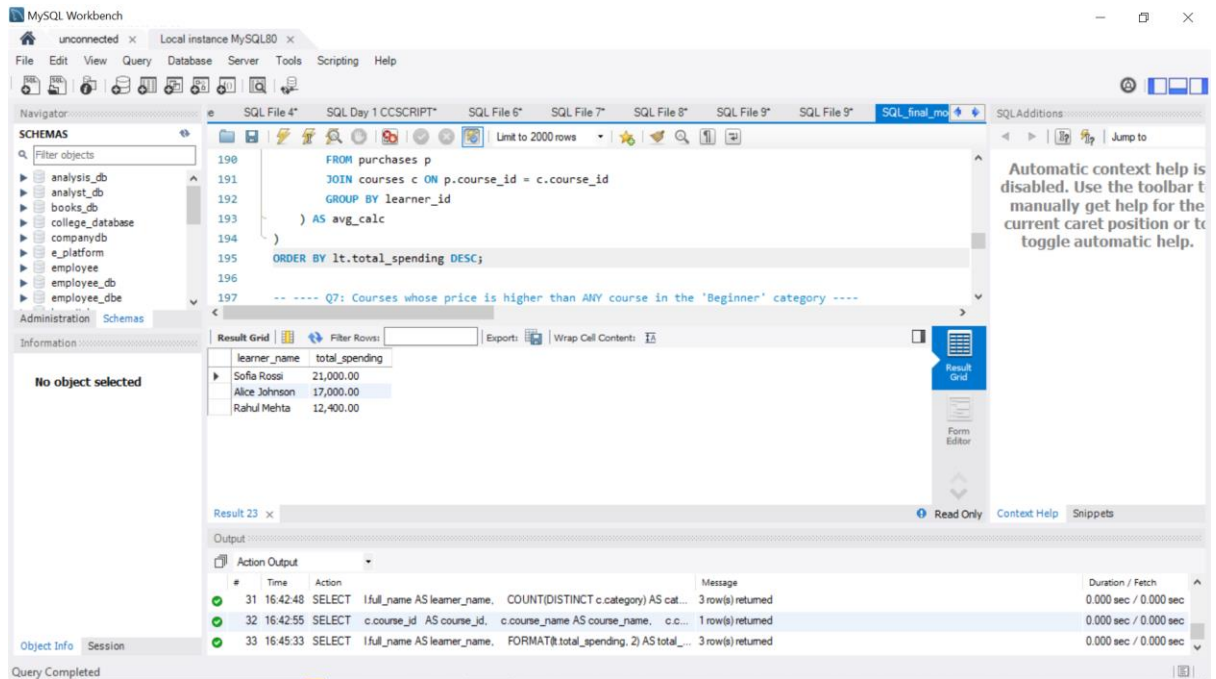
Action Output

#	Time	Action	Message	Duration / Fetch
32	16:42:55	SELECT c.course_id AS course_id, c.course_name AS course_name, c.c...	1 row(s) returned	0.000 sec / 0.000 sec
33	16:45:33	SELECT l.full_name AS learner_name, FORMAT(l.total_spending, 2) AS total...	3 row(s) returned	0.000 sec / 0.000 sec
34	16:45:41	SELECT course_name AS course_name, category AS category, FORMA...	4 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.



5. CTE, CASE, View, and NULL Handling Q9. Use a CTE to calculate total spending per learner, then: Display learners with spending above 10,000. Q10. CASE Expression Classify learners based on spending: ● Above 15,000 → “High Value”, ● 8,000–15,000 → “Medium Value”, ● Below 8,000 → “Low Value”.

Q11 . NULL Handling

- Display all courses and replace NULL purchase counts with 0 using: IFNULL() or COALESCE()

Q12 . View

- Create a view: category_performance_view ● Showing: ● Category ● Total revenue ● Number of purchases ● Average revenue per purchase

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

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Administration Schemas

Information

No object selected

SQL File 4* SQL Day 1 CCSCRIPT* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 9* SQL_final_mo

Limit to 2000 rows

271 END AS value_segment

272 FROM learner_spending

273 ORDER BY total_spending DESC;

274

275 -- ---- Q11: NULL handling - all courses, NULL purchase counts replaced with 0 ----

276 • SELECT

277 c.course_id AS course_id,

278 c.course_name AS course_name,

Result Grid

learner_name	total_spending	value_segment
Chen Wei	5,000.00	Low Value
Sofia Rossi	21,000.00	High Value
Alice Johnson	17,000.00	High Value
Rahul Mehta	12,400.00	Medium Value
John Smith	1,200.00	Low Value

Result 29 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
37	16:48:05	WITH learner_spending AS (SELECT l.learner_id, l.full_name, S...	3 row(s) returned	0.016 sec / 0.000 sec
38	16:48:32	WITH learner_spending AS (SELECT l.learner_id, l.full_name, S...	3 row(s) returned	0.000 sec / 0.000 sec
39	16:48:40	WITH learner_spending AS (SELECT l.learner_id, l.full_name, S...	5 row(s) returned	0.000 sec / 0.000 sec

Query Completed

MySQL Workbench

unconnected x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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Administration Schemas

Information

No object selected

SQL File 4* SQL Day 1 CCSCRIPT* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 9* SQL_final_mo

Limit to 2000 rows

250 WHERE total_spending > 10000

251 ORDER BY total_spending DESC;

252

253 -- ---- Q10: CASE expression to classify learners by spending tier ----

254 • WITH learner_spending AS (

255 SELECT

256 l.learner_id,

257 l.full_name,

Result Grid

learner_name	total_spending
Sofia Rossi	21,000.00
Alice Johnson	17,000.00
Rahul Mehta	12,400.00

Result 28 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
36	16:47:57	WITH learner_spending AS (SELECT l.learner_id, l.full_name, S...	3 row(s) returned	0.000 sec / 0.000 sec
37	16:48:05	WITH learner_spending AS (SELECT l.learner_id, l.full_name, S...	3 row(s) returned	0.016 sec / 0.000 sec
38	16:48:32	WITH learner_spending AS (SELECT l.learner_id, l.full_name, S...	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

MySQL Workbench

unconnected x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

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Administration Schemas

Information

No object selected

SQL File 4* SQL Day 1 CCSCRIPT* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 9* SQL_final.ms

Limit to 2000 rows

```
298 -- Query the view
299
300 • SELECT
301     category,
302     FORMAT(total_revenue, 2) AS total_revenue,
303     number_of_purchases,
304     FORMAT(avg_revenue_per_purchase, 2) AS avg_revenue_per_purchase
305 FROM category_performance_view
```

Result Grid

category	total_revenue	number_of_purchases	avg_revenue_per_purchase
Advanced	30,000.00	2	15,000.00
Intermediate	15,000.00	2	7,500.00
Beginner	11,600.00	4	2,900.00

Result 30 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
52	16:56:29	SELECT category, FORMAT(total_revenue, 2) AS total_revenue, number_o...	Error Code: 1146. Table 'elearning_platform.category_performance_view' doesn't exist	0.000 sec
53	16:58:45	CREATE OR REPLACE VIEW category_performance_view AS SELECT c.categ...	0 row(s) affected	0.031 sec
54	16:58:49	SELECT category, FORMAT(total_revenue, 2) AS total_revenue, number_o...	3 row(s) returned	0.015 sec / 0.000 sec

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.